

Project 5: Shadow Detection and Removal

Background

Shadows in pictures are known to all and very common in photography. But they will interfere the reliability of computer vision algorithms, including segmentation, object detection, tracking and so on. Therefore, shadow detection and removal are important pre-processing steps to improve the performance of these tasks.

Goal

In this project, you will implement an algorithm to detect and remove the shadows in images and produce shadow-free images. We will provide 55 images with shadows and the corresponding shadow-free images as ground truth. You need to implement the shadow removal algorithm on as many of those images as possible. The algorithm can be automatic or user-assisted. Please try to achieve the best shadow removal effect.

An example of shadow removal is provided in Figure 1. The performance of your results can be different from our example, but you need to demonstrate the advantages of your implemented algorithm in your report and presentation.



Figure 1: Example of shadow image and corresponding shadow-free image.

The images data of this project are from:

<https://drive.google.com/file/d/1W8vBRJYDG9imMgr9I2XaA13tIFIEHOjS/view>

You can add your own images if you think the provided ones are not challenging enough for your method.

Hints

You may find some hints in the following references:

[1] Liu, Feng, and Michael Gleicher. "Texture-consistent shadow removal." Computer Vision–ECCV 2008: 10th European Conference on Computer Vision, Marseille, France, October 12-18, 2008, Proceedings, Part IV 10. Springer Berlin Heidelberg, 2008.

[2] Shor, Yael, and Dani Lischinski. "The shadow meets the mask: Pyramid-based shadow removal." Computer Graphics Forum. Vol. 27. No. 2. Oxford, UK: Blackwell Publishing Ltd, 2008.

[3] Xiao, Chunxia, et al. "Efficient shadow removal using subregion matching illumination transfer." Computer Graphics Forum. Vol. 32. No. 7. 2013.

[4] Zhang, Ling, Qing Zhang, and Chunxia Xiao. "Shadow remover: Image shadow removal based on illumination recovering optimization." IEEE Transactions on Image Processing 24.11 (2015): 4623-4636.

[4] Vicente, T., Hou, L., Yu, C. P., Hoai, M., & Samaras, D. (2016). Large-Scale Training of Shadow Detectors with Noisily Annotated Shadow Examples. European Conference on Computer Vision. Springer International Publishing.

Evaluation

- a) **The code of your project: 50%.** Please ensure that your code can be run without any modification by TAs. Your code should provide the same results as in your report.
- b) **The report of this project: 30%.** You should complete your project report with template we provided. The report should contain at least these parts: a brief introduction of your project (include your motivation, background of this problem, related work of the method you adopted), the description of your method (clearly state each step and the outcome, use equations, flowchart or figures to help with the explanation when necessary), the results of your code, and discussion of your results (is your result perfect, why? If not, why is that and how to improve?).
- c) **The presentation of your project: 20%.** There will be a short presentation for about ten minutes and please prepare for it.

Please state your workload in details both in the report and in the presentation. Workload is one of the important criteria for our scoring (don't necessarily limit your report to what worked!)

Policy on Plagiarism

While you may discuss the ideas and algorithms, at NO time may you read, possess, or submit the solution code of anyone else (including people outside this course), or allow anyone else to read or possess your source code. We will detect plagiarism using automated tools and any violations will result a zero score for this assignment.